

JUDE WILLIAM COLEMAN

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EDUCATION

Trevecca Nazarene University Nashville, TN

B.S. Electrical Engineering | GPA: 3.75 / 4.00 Expected December 2027

- Relevant coursework: Embedded Systems Design, Signals & Systems, Probability, Circuits, semiconductors.....

EXPERIENCE

North Carolina Department of Adult Correction — Engineering Department Raleigh, NC

Engineering Intern, reporting to Engineering Manager

(June 2026 – Present)

- Designed and built a **Python inventory management application** (Tkinter GUI) automating asset tracking for department facilities — batch-generating serialized asset records with owner, model, and status metadata, replacing a one-by-one manual process for [X] assets.
- Implemented filename sanitization, batch file transfer, and per-record metadata generation with error handling so single failures don't halt batch operations; tool used to stand up the department's inventory management system.
- Identified the automation opportunity and delivered the tool **self-directed, without being tasked**, from requirements through deployment.
- Helped with the programming of ip based poe cameras for 2.5 full prisons

PROJECTS

Hayward Pool Heater Monitor MK5 — Junior Design (Industry-Sponsored)

(2025 – 2026)

Embedded monitoring system sponsored by Hayward Industries & Pro Charging Systems LLC

- Designed a **custom PCB in KiCad** (5 revisions) for real-time pool heater monitoring: 4-channel thermistor sensing, voltage-divider power monitoring, AMS1117-3.3 regulation, micro SD storage (**SPI**), and a **UART** docking port for an MCP39F511A power meter.
- Wrote **PIC18 firmware in embedded C** (MPLAB X): timer-interrupt timekeeping with race-safe reads, 2000-sample ADC averaging across 4 channels, thermistor resistance-to-temperature conversion, and out-of-range fault detection with safe SD shutdown logic to prevent data corruption on power loss.
- Integrated a **Raspberry Pi** data-logging layer for remote access by Hayward engineers; presented design, standards compliance (NEC, MISRA C principles), and safety analysis in a formal IEEE-style report.
- **Completed the second half of the project solo** after a team restructuring — owning PCB design, firmware, hardware testing, and documentation end-to-end.

Inventory Control System — Python / Tkinter

(2026)

- Desktop application for mass creation and organization of serialized asset records: batch record generation from pasted serial/name/model lists, status classification, integrated file transfer, and a live operation console.

TECHNICAL SKILLS

Languages: C (embedded), C++, Python

Embedded & Hardware: PIC18 microcontrollers, MPLAB X, KiCad (PCB design), Raspberry Pi, UART, SPI, ADC, thermistor sensing, power monitoring

Systems & Tools: Linux (Arch, Linux Mint, Cachy), PuTTY, TFTP, Git

Engineering Software: SolidWorks, LabVIEW